Roll No.: ....

END SEMESTER EXAMINATION JUNE 2025

Name of the Program: B.Tech

Semester: VI

Name of the Course: DevOps on Cloud

Course Code: TCS 651

Time: 3 Hours

Maximum Marks: 100

Note:

- i. **All questions are compulsory.**
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Compare and contrast Agile, DevOps, and Waterfall in terms of release frequency, automation, and feedback loops.	CO1
(b)	Describe the components of a DevOps ecosystem. Include tools used at each stage (CI, CD, Monitoring, etc.). What is the role of culture in DevOps transformation?	CO2
(c)	Analyze the difference between Docker Swarm and Kubernetes in terms of orchestration, scaling, and resilience.	CO4
Q2		
(a)	Mango Inc. is a healthcare company where you are working as a DevOps engineer. Your team uses Git and works in a remote environment. Demonstrate how Git workflows (feature branch, pull requests, etc.) help in collaborative development.	CO1
(b)	Explain how Jenkins supports Continuous Integration. Include a step-by-step flow of a Jenkins job pulling code from Git and running a build.	CO2 CO3
(c)	Describe a typical Continuous Deployment pipeline. What stages would it include? Explain with tools used.	
Q3		
(a)	Explain how Docker containers work. Show how you would containerize a simple Node.js web app.	CO4
(b)	What is a Version Control System? Describe branching and merging in Git. How do they help in collaborative development?	CO4
(c)	Explain Continuous Integration (CI). How does Jenkins facilitate CI in software projects?	CO3
Q4		
(a)	Explain the advantages of containerizing applications using Docker.	CO3
(b)	What is Selenium? Explain how it is used for automated browser testing. Describe how Selenium can be integrated with Jenkins to automate test execution.	CO4 CO5
(c)	Demonstrate step-by-step how to create a Dockerfile for a Node.js application, build the image, run the container, and connect it to a volume and network. Add relevant CLI commands.	
Q5		
(a)	Evaluate the pros and cons of rolling updates and blue-green deployments in Kubernetes. When would you prefer one over the other? Justify using a high-availability SaaS product scenario.	CO5
(b)	Compare Docker Swarm and Kubernetes in the context of cluster management, resilience, and service discovery.	CO4
(c)	Describe how Kubernetes Deployments, Pods, and ReplicaSets work together to ensure fault-tolerant applications.	CO5